MODULE 1 · CPDLC PRINCIPLES

Lesson 03: Principles of CPDLC Exchanges and Message Composition

Lesson 3 of 6 · about 7 min

LEARNING OBJECTIVES

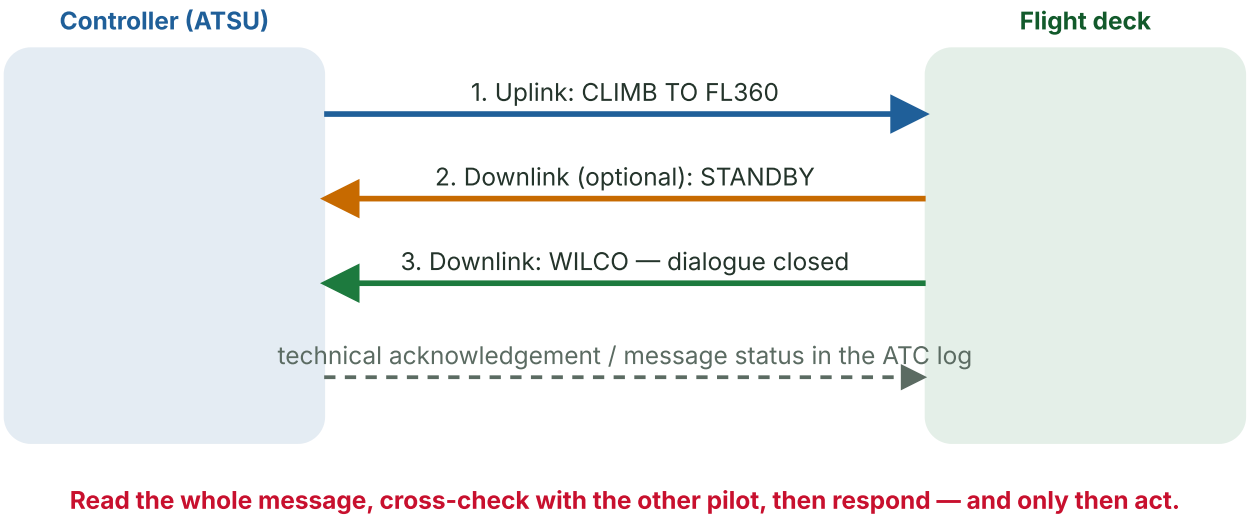
- Apply the principles for choosing voice or CPDLC.
- Understand single- and multi-element messages.
- Respond correctly to messages.



Read the whole message before responding.

Principles

- **Voice and data link co-exist.** CPDLC supplements voice; it does not replace it.
- CPDLC is used only for **non-time-critical** communications. Time-criticality depends on the traffic situation, the end-to-end performance (systems and human response times) and the recovery time.
- The choice of voice or CPDLC is at the discretion of the controller and the pilot.
- Usually, a CPDLC message is answered by CPDLC and a voice message by voice.



A typical CPDLC dialogue. The response closes the dialogue; STANDBY keeps it open.

Message composition

- Messages use a **standard format** from the ICAO message set, in plain language or abbreviations — without non-essential words.
- A message can have a **single element** (for example `CLIMB TO FL360`) or **several elements** (for example `CLIMB TO FL360, CLIMB AT 1000 FT PER MINUTE MINIMUM`). The maximum number of elements depends on the system and the airspace.
- A single response to a multi-element message applies to **all elements**. If any part cannot be complied with, respond **UNABLE** to the whole message.
- Only one open dialogue of the same type with the same aircraft is allowed at a time: horizontal profile, vertical profile, speed, or SSR code. A speed and a level dialogue can be open together; a heading and a direct-route dialogue cannot.
- Systems must handle unsupported messages correctly, normally with an error message.

Responses

Response	Meaning
WILCO	I will comply with the clearance or instruction.
UNABLE	I cannot comply (with the whole message).
STANDBY	Wait — the message is being considered; a further response will follow. The dialogue stays open.
AFFIRM / NEGATIVE	Yes / no to a question.
ROGER	Message received (for information messages).

KEY TAKEAWAYS

- Voice and CPDLC co-exist; CPDLC for non-time-critical messages only.
- One response applies to all elements of a multi-element message — UNABLE if any part cannot be done.
- Only one open dialogue of each type at a time.

KNOWLEDGE CHECK

You receive a two-element message and can comply with only one element. What do you respond?

A. WILCO

B. UNABLE for the whole message

C. ROGER

D. AFFIRM

Answer: B. If any element cannot be complied with, respond UNABLE to the whole message.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.